Final Exam

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Question 1

Consider a group of individuals where all individuals have the same utility function over consumption and labour given by: $U(c, l) = c - \frac{l^2}{2}$, where c represents consumption and l represents hours of labour supplied. Individuals work at an hourly wage of w which is taxed at a rate τ and only source of income is labour income.

(a) **Write down and draw the budget constraint faced by the individual. Solve for the individual's optimal labour supply as a function of the wage w and the tax rate τ. (5 points)**

(b) **Find out the revenue maximizing tax rate for the government. (5 points)**

(c) **Suppose that the government uses all the collected revenue to provide everyone in the economy with some universal basic income (i.e., a lump sum transfer $T > 0$ per person). How is the individual's optimal labour supply affected with this transfer? Discuss the income and substitution effects, and draw their new budget constraint. (5 points)**

Now, from here onwards, suppose there are two individuals in the economy: one earns 20 per hour and another earns 100 per hour.

(d) **Solve for each individual's utility-maximizing labour supply under the revenue maximizing tax at rate τ^* that you find in part (b), and calculate the revenue generated by the government. What is each individual's post tax earnings if the government evenly divides tax revenue to provide universal basic income? (7 points)**

(e) **Consider there has been a change in ruling government and the new government does not believe in providing universal basic income. Instead, they prefer a tax-transfer policy, as follows:**

- **There is a 100% subsidy on the first 1000 of labour income, which means that if you earn up to 1000, the government provides you with a transfer equal to your earnings.**

- **All labour income above 1000 is then taxed at rate 50%.**

Find each individual's new labour supply and after tax earnings with this changed tax-transfer scheme. (8 points)

(f) **Discuss how the substitution and income effects influence their optimal labour supply relative to the previous tax scheme. (5 points)**

(g) **Is the low wage individual better off under the tax scheme from question (e) or under the tax scheme from question (d)? Is the high wage individual better off under the tax scheme from question (e) or under the tax scheme from question (d)? Is society better off under the tax scheme from question (e) or under the tax scheme from question (d)? (5 points)**

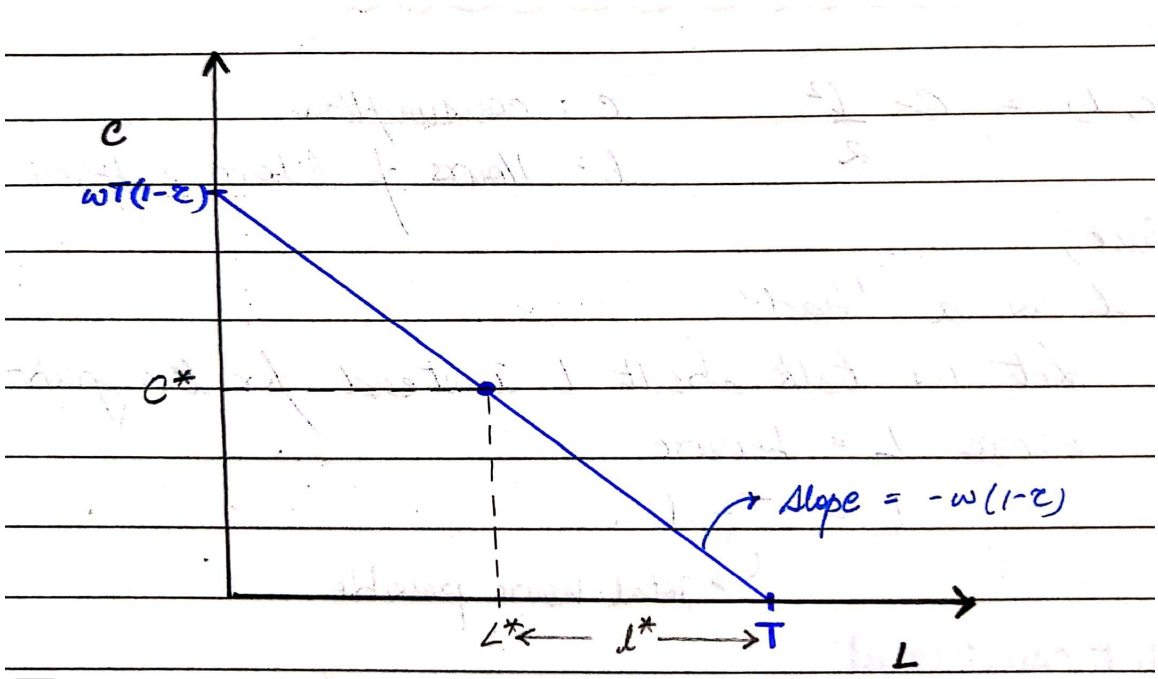
Answer

(a) **Budget Constraint**

The individual has no endowment, the only income they get is their labour income. Clearly, the total amount of income earned from labour is wl. However, their consumption is not equal to this since they have to pay a tax at the rate of τ. So, they end up paying $wl\tau$ as tax and hence, since net income has to be at least as much as the consumption. So their budget constraint is:

$$\boldsymbol{c \leq wl(1-\tau)}$$

For the graph, we can use L, which stands for 'leisure' and can mathematically be expressed as $T - l$, where T is the maximum number of hours a person can work. We are using L because l can be thought of as a 'bad'. The budget constraint graph is as follows:



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Optimal labour supply

$$max_{c,l} U(c,l) = c - \frac{l^2}{2}$$

such that

$$c \leq wl(1-\tau)$$

Clearly, utility will be maximised when

$$c = wl(1-\tau)$$

Plugging this into the objective function, we get

$$U(c,l) = wl(1-\tau) - \frac{l^2}{2}$$

Differentiating w.r.t l:

$$\frac{dU}{dl} = w(1-\tau) - \frac{2l}{2}$$

$$\implies \frac{dU}{dl} = w(1-\tau) - l$$

First order condition:

$$w(1-\tau) - l = 0$$

$$\implies l = w(1-\tau)$$

Checking second order condition to ensure maxima:

$$\frac{d^2U}{dl^2} = -1$$

$$\implies \frac{d^2U}{dl^2} < 0$$

$\implies$ Hence, the optimal labour supply is $\boldsymbol{l^* = w(1-\tau)}$.

(b)

$$max_\tau T = \tau w l$$

We know that

$$l^* = w(1-\tau)$$

Therefore

$$\tau w l = \tau w^2 (1-\tau)$$

Differentiating w.r.t τ:

$$\frac{dT}{d\tau} = w^2(1-\tau) - \tau w^2$$

$$\implies \frac{dT}{d\tau} = w^2 - 2\tau w^2$$

First order condition:

$$w^2 - 2\tau w^2 = 0$$

$$\implies w^2 = 2\tau w^2$$

Since $w \neq 0$

$$2\tau = 1$$

$$\implies \tau = 0.5 = 50\%$$

$\implies$ Hence, the tax rate that maximises government revenue is **50**%.

(c) The new budget constraint is as follows:

$$c \leq wl(1-\tau) + T$$

$$max_{c,l} U(c,l) = c - \frac{l^2}{2}$$

such that

$$c \leq wl(1-\tau) + T$$

$$U(c,l) = wl(1-\tau) + T - \frac{l^2}{2}$$

Differentiating w.r.t l:

$$\frac{dU}{dl} = w(1-\tau) - \frac{2l}{2}$$

$$\implies \frac{dU}{dl} = w(1-\tau) - l$$

First order condition:

$$w(1-\tau) - l = 0$$
$$\implies l = w(1-\tau)$$

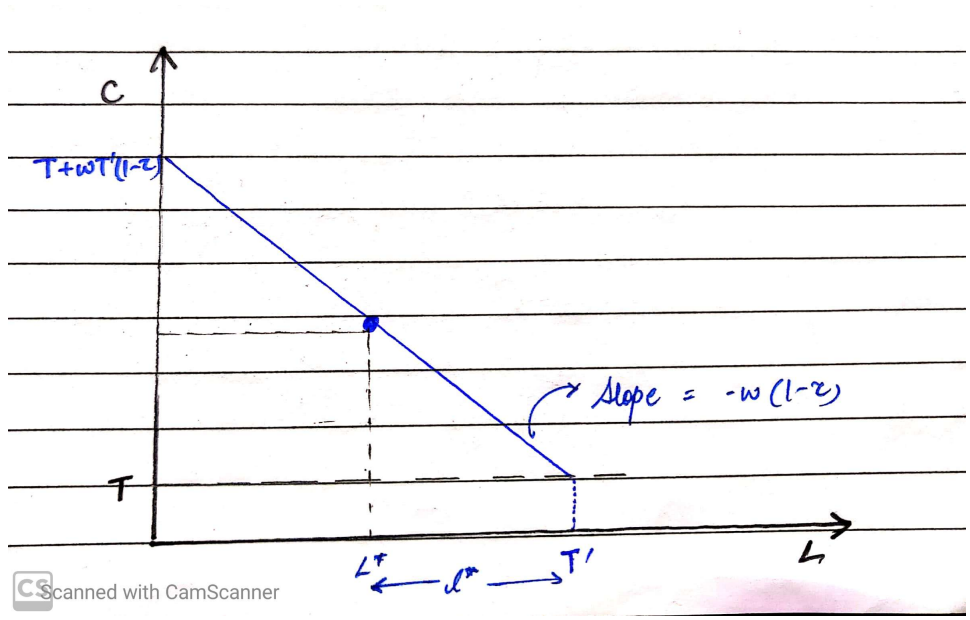
Checking second order condition to ensure maxima:

$$\frac{d^2U}{dl^2} = -1$$
$$\implies \frac{d^2U}{dl^2} < 0$$

$\implies$ Hence, the optimal labour supply is $\boldsymbol{l^* = w(1-\tau)}$.

This implies that there is no change.

The new budget constraint is shown below:



We see that the function is quasilinear for labour (leisure) supply and hence income effect is 0. Since the slope did not change (as seen in the figures), substitution effect is also zero.

(d) We know that $l^* = w(1-\tau)$ At the revenue maximising level

$$\tau = 0.5$$
$$\implies l^* = 0.5w$$

Revenue generated by the government:

$$R = \tau w l$$

$$\implies R = (0.5)(20)(0.5)(20) + (0.5)(100)(0.5)(100)$$
$$\implies R = 100 + 2500$$
$$\implies R = 2600$$

Post tax earnings:

Person1: $(0.5)(20)(0.5)(20)+1300 = 1400$ Person1: $(0.5)(100)(0.5)(100)+1300 = 3800$

(e) For $wl \leq 1000$, $l = 2w$. For $wl > 1000$, $l = 0.5w$.

For person 1, $l = 2w = 40$ hours.

After tax earning $= 2wl = 1600$

For person 2, $l = w/2 = 50$ hours.

After tax earning $= 2000 + 2000 = 4000$

(f) Quasilinear, so no income effect, for individual one, S.E. is positive as opportunity cost of leisure goes up and negative for individual two as opportunity cost for leisure goes down.

(g) Low wage individual is definitely better off and even the high wage individual is better off slightly.

Society as a whole is also better off.

Question 2

Suppose the government of India has asked you to re-evaluate the costs and benefits of various consumption and income tax proposals that his tax panel has made. In order to do that, you consider a two period model where individuals earn labour income Y from working in period 1 and do not work in period 2 (retirement). Individuals choose how much to consume in each period. Savings in period 1 earn an interest rate $r > 0$. Let c_1 denote consumption in period 1 and c_2 denote consumption in period 2.

(a) Write the individual's budget constraint in an economy without taxes. (3 points)

(b) Write the budget constraint where both labour and capital income are taxed at rate t, assuming there is only one tax bracket and there are no tax-exempt savings accounts. (3 points)

(c) Write the budget constraint with a consumption tax rate τ and no comprehensive income tax. (4 points)

(d) **The tax panel claims that exempting capital income from the income tax while retaining the income tax on labour income is equivalent to shifting to a consumption tax system. Prove this algebraically using the budget constraints in parts (a) and (b), still assuming one income tax bracket. (5 points)**

(e) **Suppose that individuals have a utility function $u(c_1, c_2) = c_1^{0.5} + (\frac{c_2}{1+r})^{0.5}$. Show that a consumption tax rate τ does not distort consumption choices. (5 points)**

Answer

(a) Clearly, the budget constraint is as follows:

$$c_1 + \frac{c_2}{1+r} \leq Y$$

(b) Since there is a tax t on both labour and capital income, the budget constraint will now look like this:

$$c_1 + \frac{c_2}{1+r(1-t)} \leq Y(1-t)$$

Hence, we see that the tax distorts both labour supply and savings. It imposes a 'double' tax, both on earnings and savings.

(c) The budget constraint with a consumption tax, and no comprehensive income tax will look like this:

$$(1+\tau)(c_1 + \frac{c_2}{1+r}) \leq Y$$

(d) 1. Budget constraint with only consumption tax

$$(1+\tau)(c_1 + \frac{c_2}{1+r}) \leq Y$$

2. Budget constraint with only labour income tax

$$c_1 + \frac{c_2}{1+r} \leq Y(1-t)$$

Now, equation 2 can be re-written as:

$$\frac{1}{1-t}(c_1 + \frac{c_2}{1+r}) \leq Y$$

It is evident that this is the same as equation 1 when

$$\frac{1}{1-t} = 1+\tau$$

$$\implies \tau = \frac{1}{1-t} - 1$$

The panellists are right, a labour income tax t while exempting capital income is the same as a consumption tax $\frac{1}{1-t} - 1$.

(e) Let us first look at the optimisation without the tax

$$max_{c_1,c_2} u(c_1, c_2) = c_1^{0.5} + (\frac{c_2}{1+r})^{0.5}$$

such that

$$c_1 + \frac{c_2}{1+r} \leq Y$$

At the optimal point

$$c_1 + \frac{c_2}{1+r} \leq Y$$
$$\implies \frac{c_2}{1+r} \leq Y - c_1$$

Substitution this in the objective function, we get

$$max_{c_1} (c_1)^{0.5} + (Y - c_1)^{0.5}$$

Differentiating w.r.t. c_1:

$$\frac{du}{dc_1} = \frac{c_1^{-0.5}}{2} - \frac{(Y - c_1)^{-0.5}}{2}$$

First order condition:

$$\frac{c_1^{-0.5}}{2} = \frac{(Y - c_1)^{-0.5}}{2}$$
$$\implies c_1^{-0.5} = (Y - c_1)^{-0.5}$$

Squaring and inverting:

$$\implies c_1 = Y - c_1$$
$$\implies 2c_1 = Y$$

$$\implies \boldsymbol{c_1 = \frac{Y}{2}}$$

From budget constraint:

$$c_1 + \frac{c_2}{1+r} = Y$$
$$\implies \frac{c_2}{1+r} = \frac{Y}{2}$$

$$\implies \boldsymbol{c_2 = \frac{Y(1+r)}{2}}$$

Calculating the ratio

$$\boldsymbol{\frac{c_2}{c_1} = 1 + r}$$

Introducing a consumption tax τ

$$max_{c_1,c_2} u(c_1, c_2) = c_1^{0.5} + (\frac{c_2}{1+r})^{0.5}$$

The budget constraint changes to:

$$(1+\tau)(c_1 + \frac{c_2}{1+r}) \leq Y$$

At the optimal point

$$(1+\tau)(c_1 + \frac{c_2}{1+r}) = Y$$

$$\implies \frac{c_2}{1+r} = \frac{Y}{1+\tau} - c_1$$

Substitution this in the objective function, we get

$$max_{c_1}(c_1)^{0.5} + (\frac{Y}{1+\tau} - c_1)^{0.5}$$

Differentiating w.r.t. c_1:

$$\frac{du}{dc_1} = \frac{c_1^{-0.5}}{2} - \frac{(\frac{Y}{1+\tau} - c_1)^{-0.5}}{2}$$

First order condition:

$$c_1^{-0.5} = (\frac{Y}{1+\tau} - c_1)^{-0.5}$$

Squaring and inverting:

$$c_1 = \frac{Y}{1+\tau} - c_1$$

$$\implies 2c_1 = \frac{Y}{1+\tau}$$

$$\implies \boldsymbol{c_1 = \frac{Y}{2(1+\tau)}}$$

From budget constraint:

$$(1+\tau)(c_1 + \frac{c_2}{1+r}) = Y$$

$$\implies \frac{Y}{2(1+\tau)} + \frac{c_2}{1+r} = \frac{Y}{(1+\tau)}$$

$$\implies \frac{Y}{2(1+\tau)} + \frac{c_2}{1+r} = \frac{Y}{(1+\tau)}$$

$$\implies \frac{c_2}{1+r} = \frac{Y}{2(1+\tau)}$$

$$\implies \boldsymbol{c_2 = \frac{Y(1+r)}{2(1+\tau)}}$$

Calculating the ratio

$$\boldsymbol{\frac{c_2}{c_1} = 1 + r}$$

Hence, we see that a consumption tax rate τ does not distort consumption choices.